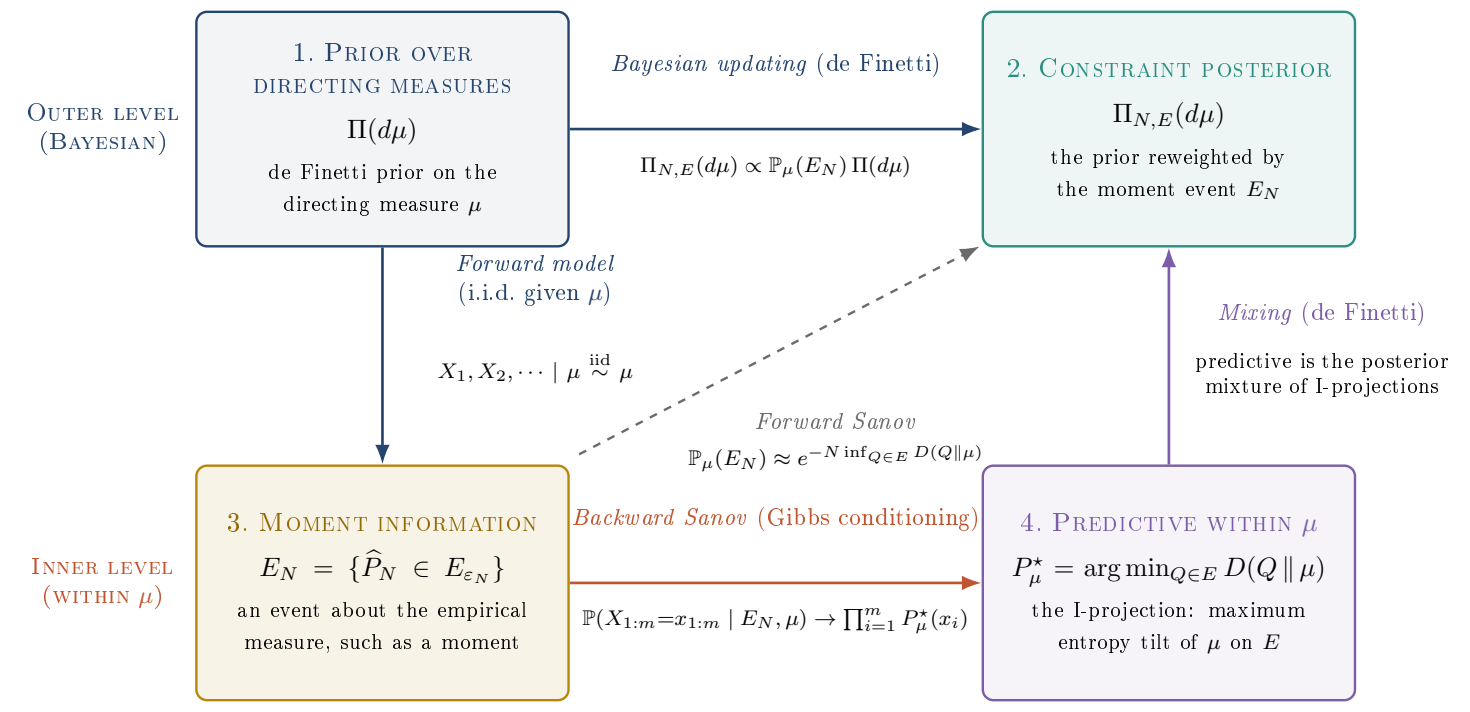

Outer level
(Bayesian)
1. Prior over directing measures
$\Pi(d\mu)$
de Finetti prior on the directing measure μ
Bayesian updating (de Finetti)
$\Pi_{N,E}(d\mu) \propto \mathbb{P}_\mu(E_N)\,\Pi(d\mu)$
2. Constraint posterior
$\Pi_{N,E}(d\mu)$
the prior reweighted by the moment event E_N
Forward model
(i.i.d. given μ)
$X_1, X_2, \cdots \mid \mu \overset{\text{iid}}{\sim} \mu$
Forward Sanov
$\mathbb{P}_\mu(E_N) \approx e^{-N \inf_{Q\in E} D(Q\|\mu)}$
Mixing (de Finetti)
predictive is the posterior mixture of I-projections
Inner level
(within μ)
3. Moment information
$E_N = \{\widehat{P}_N \in E_{\varepsilon_N}\}$
an event about the empirical measure, such as a moment
Backward Sanov (Gibbs conditioning)
$\mathbb{P}(X_{1:m}{=}x_{1:m} \mid E_N, \mu) \to \prod_{i=1}^m P^\star_\mu(x_i)$
4. Predictive within μ
$P^\star_\mu = \arg\min_{Q\in E} D(Q \,\|\, \mu)$
the I-projection: maximum entropy tilt of μ on E